TOPIC 1 OF 77

FIRST - ANALYZE THE ENTIRE PROJECT

Requirements understanding

Inspect the real repository before making any documentation claim. Identify languages, frameworks, models, tools, APIs, state, memory, tests, deployment files, and missing features.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Repository -> dependencies/config -> entry points -> agent components -> APIs/tools -> tests -> evidence-backed docs
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 2 OF 77

PROJECT OVERVIEW

Requirements understanding

Explain the project purpose, problem, inputs, outputs, capabilities, and value at beginner, technical, and interview levels.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Problem -> user input -> agent capability -> output -> beginner explanation -> technical explanation
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 3 OF 77

WHAT IS AN AI AGENT?

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 4 OF 77

AI AGENT VS NORMAL LLM

Requirements understanding

Compare the two concepts using definitions, observable behavior, an example, and a practical rule of thumb. Keep distinctions precise.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Concept A -> definition -> Concept B -> definition -> difference -> example -> rule of thumb
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 5 OF 77

AI AGENT ARCHITECTURE

Requirements understanding

Show how the parts connect as one system: user, frontend, API, agent, model, tools, external services, state, and response.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
User -> frontend -> backend API -> agent <-> LLM -> tool/service -> observation -> final response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

COMPLETE AGENT PIPELINE

Requirements understanding

Show how the parts connect as one system: user, frontend, API, agent, model, tools, external services, state, and response.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
User -> frontend -> backend API -> agent <-> LLM -> tool/service -> observation -> final response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 7 OF 77

WHAT MAKES THIS PROJECT AN AGENT?

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 8 OF 77

COMPLETE USER REQUEST FLOW

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

LLM

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 10 OF 77

AGENT FRAMEWORK

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 11 OF 77

AGENT CORE

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOOLS

Requirements understanding

Document tool purpose, schema, arguments, execution, result handling, permissions, and failure behavior. Only describe tools confirmed in code.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Tool schema -> model selection -> arguments -> validation -> service call -> result -> agent
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOOL CALLING

Requirements understanding

Document tool purpose, schema, arguments, execution, result handling, permissions, and failure behavior. Only describe tools confirmed in code.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Tool schema -> model selection -> arguments -> validation -> service call -> result -> agent
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

FUNCTION CALLING

Requirements understanding

Document tool purpose, schema, arguments, execution, result handling, permissions, and failure behavior. Only describe tools confirmed in code.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Tool schema -> model selection -> arguments -> validation -> service call -> result -> agent
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 15 OF 77

AGENT DECISION LOOP

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 16 OF 77

SINGLE-STEP AGENT

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 17 OF 77

MULTI-STEP AGENT

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

PLANNING

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

MEMORY

Requirements understanding

Explain what information is carried between steps, where it is stored, how it is retrieved or updated, and whether it persists across sessions.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Input/history/state -> context assembly -> model/tool use -> state or memory update -> response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

AGENT STATE

Requirements understanding

Explain what information is carried between steps, where it is stored, how it is retrieved or updated, and whether it persists across sessions.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Input/history/state -> context assembly -> model/tool use -> state or memory update -> response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 21 OF 77

WORKFLOW / GRAPH

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

STOP CONDITION

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

MAXIMUM ITERATIONS

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

PROMPTS

Requirements understanding

Identify system, user, agent, tool, and output instructions. Explain their purpose without exposing secrets or private chain-of-thought.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Role + rules + tools + context + output contract -> model request -> text or tool call
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

SYSTEM PROMPT

Requirements understanding

Identify system, user, agent, tool, and output instructions. Explain their purpose without exposing secrets or private chain-of-thought.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Role + rules + tools + context + output contract -> model request -> text or tool call
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 26 OF 77

TOOL DESCRIPTIONS

Requirements understanding

Document tool purpose, schema, arguments, execution, result handling, permissions, and failure behavior. Only describe tools confirmed in code.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Tool schema -> model selection -> arguments -> validation -> service call -> result -> agent
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

CONTEXT

Requirements understanding

Explain what information is carried between steps, where it is stored, how it is retrieved or updated, and whether it persists across sessions.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Input/history/state -> context assembly -> model/tool use -> state or memory update -> response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 28 OF 77

RAG + AI AGENT

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

AGENT + API

Requirements understanding

Document actual HTTP methods, paths, headers, request and response bodies, backend processing, external calls, and errors.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
HTTP request -> route -> validation -> controller/service -> agent/model/tool -> response or error
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 30 OF 77

API DOCUMENTATION

Requirements understanding

Document actual HTTP methods, paths, headers, request and response bodies, backend processing, external calls, and errors.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
HTTP request -> route -> validation -> controller/service -> agent/model/tool -> response or error
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 31 OF 77

COMPLETE REQUEST -> RESPONSE FLOW

Requirements understanding

Show how the parts connect as one system: user, frontend, API, agent, model, tools, external services, state, and response.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
User -> frontend -> backend API -> agent <-> LLM -> tool/service -> observation -> final response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

FRONTEND FLOW

Requirements understanding

Map pages and components from user input through submit, loading, API communication, error handling, and response rendering.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Input -> submit -> loading -> API -> response/error -> render/history
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

BACKEND FLOW

Requirements understanding

Trace the backend route, controller/service, agent, model, tools, result normalization, and HTTP response using actual files.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Request -> route -> service -> agent -> model/tool -> serializer -> response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 34 OF 77

FILE-BY-FILE EXPLANATION

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

FUNCTION-BY-FUNCTION EXPLANATION

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 36 OF 77

ENVIRONMENT VARIABLES

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

LOCAL SETUP

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

COMPLETE REAL-WORLD EXAMPLE

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 39 OF 77

AGENT OBSERVATION

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

ERROR HANDLING

Requirements understanding

Describe the symptom, likely cause, detection point, handling policy, user-safe response, and verification step.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Failure -> detection -> retry/fallback/stop decision -> log/metric -> safe response -> retest
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOOL FAILURE

Requirements understanding

Document tool purpose, schema, arguments, execution, result handling, permissions, and failure behavior. Only describe tools confirmed in code.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Tool schema -> model selection -> arguments -> validation -> service call -> result -> agent
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

SECURITY

Requirements understanding

Analyze risk at every boundary. Separate current controls from production recommendations, and apply least privilege, validation, approval, and safe logging.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Identity -> authorization -> agent boundary -> permission check -> tool/external system
-> audit-safe result
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

PROMPT INJECTION

Requirements understanding

Identify system, user, agent, tool, and output instructions. Explain their purpose without exposing secrets or private chain-of-thought.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Role + rules + tools + context + output contract -> model request -> text or tool call
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOOL SECURITY

Requirements understanding

Document tool purpose, schema, arguments, execution, result handling, permissions, and failure behavior. Only describe tools confirmed in code.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Tool schema -> model selection -> arguments -> validation -> service call -> result -> agent
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

HUMAN-IN-THE-LOOP

Requirements understanding

Analyze risk at every boundary. Separate current controls from production recommendations, and apply least privilege, validation, approval, and safe logging.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Identity -> authorization -> agent boundary -> permission check -> tool/external system
-> audit-safe result
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

MEMORY AND PRIVACY

Requirements understanding

Explain what information is carried between steps, where it is stored, how it is retrieved or updated, and whether it persists across sessions.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Input/history/state -> context assembly -> model/tool use -> state or memory update -> response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

AGENT VS RAG

Requirements understanding

Compare the two concepts using definitions, observable behavior, an example, and a practical rule of thumb. Keep distinctions precise.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Concept A -> definition -> Concept B -> definition -> difference -> example -> rule of thumb
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

AGENT VS CHATBOT

Requirements understanding

Compare the two concepts using definitions, observable behavior, an example, and a practical rule of thumb. Keep distinctions precise.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Concept A -> definition -> Concept B -> definition -> difference -> example -> rule of thumb
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

AGENT VS AUTOMATION

Requirements understanding

Compare the two concepts using definitions, observable behavior, an example, and a practical rule of thumb. Keep distinctions precise.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Concept A -> definition -> Concept B -> definition -> difference -> example -> rule of thumb
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 50 OF 77

SINGLE AGENT VS MULTI-AGENT

Requirements understanding

Compare the two concepts using definitions, observable behavior, an example, and a practical rule of thumb. Keep distinctions precise.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Concept A -> definition -> Concept B -> definition -> difference -> example -> rule of thumb
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

PRODUCTION ARCHITECTURE

Requirements understanding

Show how the parts connect as one system: user, frontend, API, agent, model, tools, external services, state, and response.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
User -> frontend -> backend API -> agent <-> LLM -> tool/service -> observation -> final response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

SCALABILITY

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

PERFORMANCE

Requirements understanding

Translate the simple project into reliable operations: deployment, security, monitoring, budgets, latency control, scaling, and recovery.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Build/configure -> deploy -> health check -> trace/measure -> scale or remediate
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

COST

Requirements understanding

Translate the simple project into reliable operations: deployment, security, monitoring, budgets, latency control, scaling, and recovery.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Build/configure -> deploy -> health check -> trace/measure -> scale or remediate
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

OBSERVABILITY

Requirements understanding

Translate the simple project into reliable operations: deployment, security, monitoring, budgets, latency control, scaling, and recovery.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Build/configure -> deploy -> health check -> trace/measure -> scale or remediate
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TESTING

Requirements understanding

Define tests and measurements for correctness, tool selection, argument validity, safety, latency, cost, and end-to-end task success.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Test task -> expected behavior -> run -> score -> analyze failures -> improve -> regression check
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

AGENT EVALUATION

Requirements understanding

Define tests and measurements for correctness, tool selection, argument validity, safety, latency, cost, and end-to-end task success.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Test task -> expected behavior -> run -> score -> analyze failures -> improve -> regression check
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

DEBUGGING GUIDE

Requirements understanding

Describe the symptom, likely cause, detection point, handling policy, user-safe response, and verification step.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Failure -> detection -> retry/fallback/stop decision -> log/metric -> safe response -> retest
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

DEPLOYMENT

Requirements understanding

Translate the simple project into reliable operations: deployment, security, monitoring, budgets, latency control, scaling, and recovery.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Build/configure -> deploy -> health check -> trace/measure -> scale or remediate
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

DOCKER

Requirements understanding

Translate the simple project into reliable operations: deployment, security, monitoring, budgets, latency control, scaling, and recovery.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Build/configure -> deploy -> health check -> trace/measure -> scale or remediate
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

TOPIC 61 OF 77

BIG PROJECT IMPLEMENTATION

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

ADVANCED AGENT FEATURES

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

DETERMINISTIC WORKFLOW VS AUTONOMOUS AGENT

Requirements understanding

Compare the two concepts using definitions, observable behavior, an example, and a practical rule of thumb. Keep distinctions precise.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Concept A -> definition -> Concept B -> definition -> difference -> example -> rule of thumb
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

COMPLETE END-TO-END FLOW

Requirements understanding

Show how the parts connect as one system: user, frontend, API, agent, model, tools, external services, state, and response.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
User -> frontend -> backend API -> agent <-> LLM -> tool/service -> observation -> final response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

COMPLETE REQUEST/RESPONSE EXAMPLE

Requirements understanding

Show how the parts connect as one system: user, frontend, API, agent, model, tools, external services, state, and response.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
User -> frontend -> backend API -> agent <-> LLM -> tool/service -> observation -> final response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

SECURITY ARCHITECTURE

Requirements understanding

Show how the parts connect as one system: user, frontend, API, agent, model, tools, external services, state, and response.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
User -> frontend -> backend API -> agent <-> LLM -> tool/service -> observation -> final response
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

INTERVIEW QUESTIONS

Requirements understanding

Prepare a technically correct explanation based on verified project facts, with a clear problem, solution, architecture, flow, and tradeoffs.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Problem -> solution -> stack -> agent/model -> tools -> decision flow -> result -> tradeoffs
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

2-MINUTE PROJECT EXPLANATION

Requirements understanding

Prepare a technically correct explanation based on verified project facts, with a clear problem, solution, architecture, flow, and tradeoffs.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Problem -> solution -> stack -> agent/model -> tools -> decision flow -> result -> tradeoffs
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

5-MINUTE PROJECT EXPLANATION

Requirements understanding

Prepare a technically correct explanation based on verified project facts, with a clear problem, solution, architecture, flow, and tradeoffs.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Problem -> solution -> stack -> agent/model -> tools -> decision flow -> result -> tradeoffs
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

BEGINNER GLOSSARY

Requirements understanding

Turn the requirements into a structured learning artifact that is easy to navigate, beginner-friendly, technically accurate, and reviewable.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Source requirements -> focused topics/files -> master narrative -> render -> inspect -> deliver
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

COMMON CONFUSIONS

Requirements understanding

Compare the two concepts using definitions, observable behavior, an example, and a practical rule of thumb. Keep distinctions precise.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Concept A -> definition -> Concept B -> definition -> difference -> example -> rule of thumb
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

LEARNING ROADMAP

Requirements understanding

Turn the requirements into a structured learning artifact that is easy to navigate, beginner-friendly, technically accurate, and reviewable.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Source requirements -> focused topics/files -> master narrative -> render -> inspect -> deliver
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

DOCUMENTATION FILE STRUCTURE

Requirements understanding

Turn the requirements into a structured learning artifact that is easy to navigate, beginner-friendly, technically accurate, and reviewable.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Source requirements -> focused topics/files -> master narrative -> render -> inspect -> deliver
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

MASTER DOCUMENT

Requirements understanding

Turn the requirements into a structured learning artifact that is easy to navigate, beginner-friendly, technically accurate, and reviewable.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Source requirements -> focused topics/files -> master narrative -> render -> inspect -> deliver
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

PDF DOCUMENT

Requirements understanding

Turn the requirements into a structured learning artifact that is easy to navigate, beginner-friendly, technically accurate, and reviewable.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Source requirements -> focused topics/files -> master narrative -> render -> inspect -> deliver
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

FINAL QUALITY CHECK

Requirements understanding

Explain this topic from beginner level to implementation level, then separate what exists in the repository from what is only recommended for production.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Requirement -> repository evidence -> implementation explanation -> production
recommendation -> verification
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.

MOST IMPORTANT RULES

Requirements understanding

Apply the non-negotiable rules: inspect first, never invent implementation details, label missing features, separate current code from recommendations, and protect secrets.

Expected documentation output

Use evidence from actual source code, configuration, dependencies, routes, prompts, state, tools, tests, and runtime behavior. If absent, write **NOT IMPLEMENTED IN THIS PROJECT**. If it is future advice, write **PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED**.

Workflow

```
Inspect -> document evidence -> label unknowns -> protect secrets -> verify -> deliver
```

Accuracy checkpoint

Never infer implementation from a common AI-agent pattern. Verify it in the repository and do not expose credentials, private information, or hidden chain-of-thought.